

Dynamic Segmentation using Otsu's Thresholding

GNR 602

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Problem Statement

Segment an image by dynamic thresholding approach, with Otsu thresholding as the basis thresholding method.

Methodology - Otsu Thresholding

- The method works by considering all possible threshold values and selecting the one that minimizes the within-class variance.

$$\sigma_w^2(t) = \omega_0(t)\sigma_0^2(t) + \omega_1(t)\sigma_1^2(t)$$

$$\omega_0(t) = \sum_{i=0}^{t-1} p(i) \quad \omega_1(t) = \sum_{i=t}^{L-1} p(i)$$

- It is a global thresholding technique, but it is useful in cases where the image histogram is bimodal.



Methodology - Otsu Thresholding

Steps:

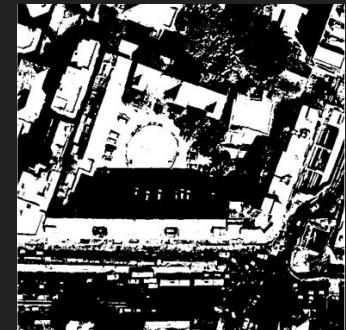
1. Compute the histogram of the image.
2. Compute the total mean level of the image.
3. For each possible threshold T , compute the class probabilities and means.
4. Compute the between-class variance for each threshold.
5. Select the threshold T that maximizes the between-class variance.

Issues with Global thresholding

- Global thresholding fails in the areas of uneven illumination, misclassifying the pixels in darker or brighter regions
- A single global value can't separate foreground/background well when their intensities are similar or overlap
- A significant amount of noise or large artifacts can distort the global histogram leading to inaccurate threshold value for the image
- In the example shown, the otsu method is not able to properly threshold the areas in the shadowed part



Otsu
Thresholding



Solution

- Instead of one global cut-off, compute Otsu's threshold within a sliding window around each pixel, so bright and dark regions are segmented appropriately.
- For every pixel, extract its local block (e.g. 35×35), run Otsu thresholding on that block, then binarize the center pixel – this handles uneven illumination.
- Because thresholds adjust to local intensity distributions, fine structures and shadowed areas become much clearer compared to global thresholding.



Dynamic
Thresholding



Results

- A global, image-wide threshold leaves runway markings and edges poorly defined against the surroundings
- Computing Otsu's threshold dynamically in a sliding window around each pixel sharply delineates runway features, resulting in a much clearer segmentation



Original Image

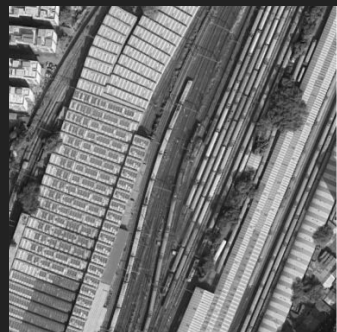


Global Otsu thresholding



Dynamic Otsu threshold

Results

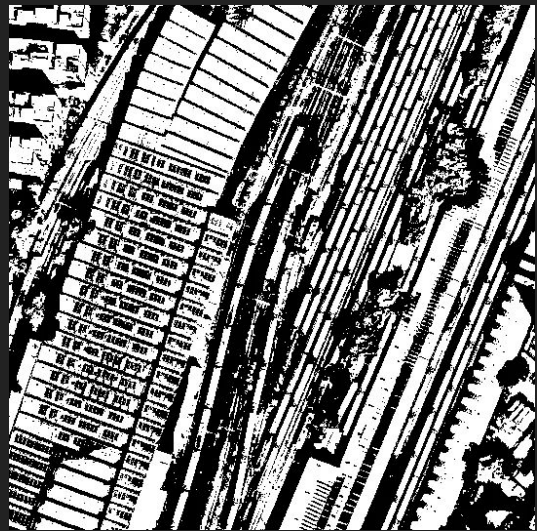


Original Image

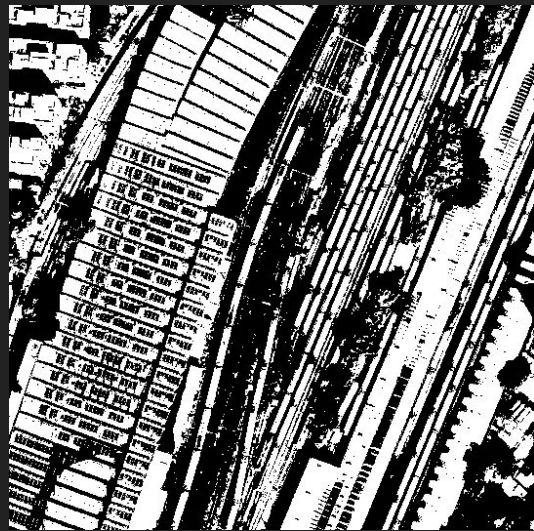
Global Otsu thresholding

Dynamic Otsu threshold

Block Size effect



Block Size: 35



Block Size: 51



Block Size: 71

THANK YOU